

SpaceX Falcon 9 First Stage Landing Prediction

Data Science Capstone Project Report

Author: Itzraj786iul

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GitHub Repository URL: https://github.com/Itzraj786iul/Spacex_Project

Contains all completed notebooks and Python files

GitHub Repository Including All Completed Notebooks and Python Files

Repository URL: https://github.com/ltzraj786iul/Spacex_Project

Completed notebooks and Python files in this repository:

1. 01-data-collection-api.ipynb - SpaceX REST API data collection
2. 01-data-collection-webscraping.ipynb - Wikipedia web scraping
3. 01-data-wrangling.ipynb - data wrangling and Class labels
4. 02-eda-sql.ipynb - EDA with SQL queries
5. 02-eda-dataviz.ipynb - EDA with data visualization
6. 03-folium-map.ipynb - interactive Folium maps
7. 03-dash-app.py - Plotly Dash dashboard application
8. 04-predictive-models.ipynb - classification predictive analysis

This PDF is the final presentation: Data Science Capstone Project Report.pdf

Executive Summary

Summary of methodologies:

- Data collection via SpaceX REST API and Wikipedia web scraping
- Data wrangling to create binary Class labels for landing success/failure
- Exploratory data analysis with SQL (SQLite) and Seaborn visualizations
- Interactive visual analytics with Folium maps and Plotly Dash
- Predictive analysis using Logistic Regression, SVM, Decision Tree, and KNN

Summary of all results:

- Success rate improved over time across launch factors
- KSC LC-39A has the highest launch success rate of all sites
- ES-L1, SSO, HEO, and GEO orbits show the highest success rates
- Heavier payloads had higher early failure rates, improving later
- DecisionTreeClassifier is the most accurate predictive model

Introduction

Project background and context:

- SpaceX Falcon 9 launches cost about \$62 million versus \$165 million+ for others
- Much of the savings comes from reusing the first stage after a successful landing
- Predicting first-stage landing success helps estimate true launch cost
- This information supports competitive bidding against SpaceX

Problems we want to find answers:

1. Which features predict successful Falcon 9 first-stage landing?
2. Which launch sites and orbits have the best success rates?
3. How do payload mass and yearly trends affect landing outcomes?
4. What geographic patterns exist around launch sites?
5. Which classification model best predicts landing success?

Data Collection Methodology

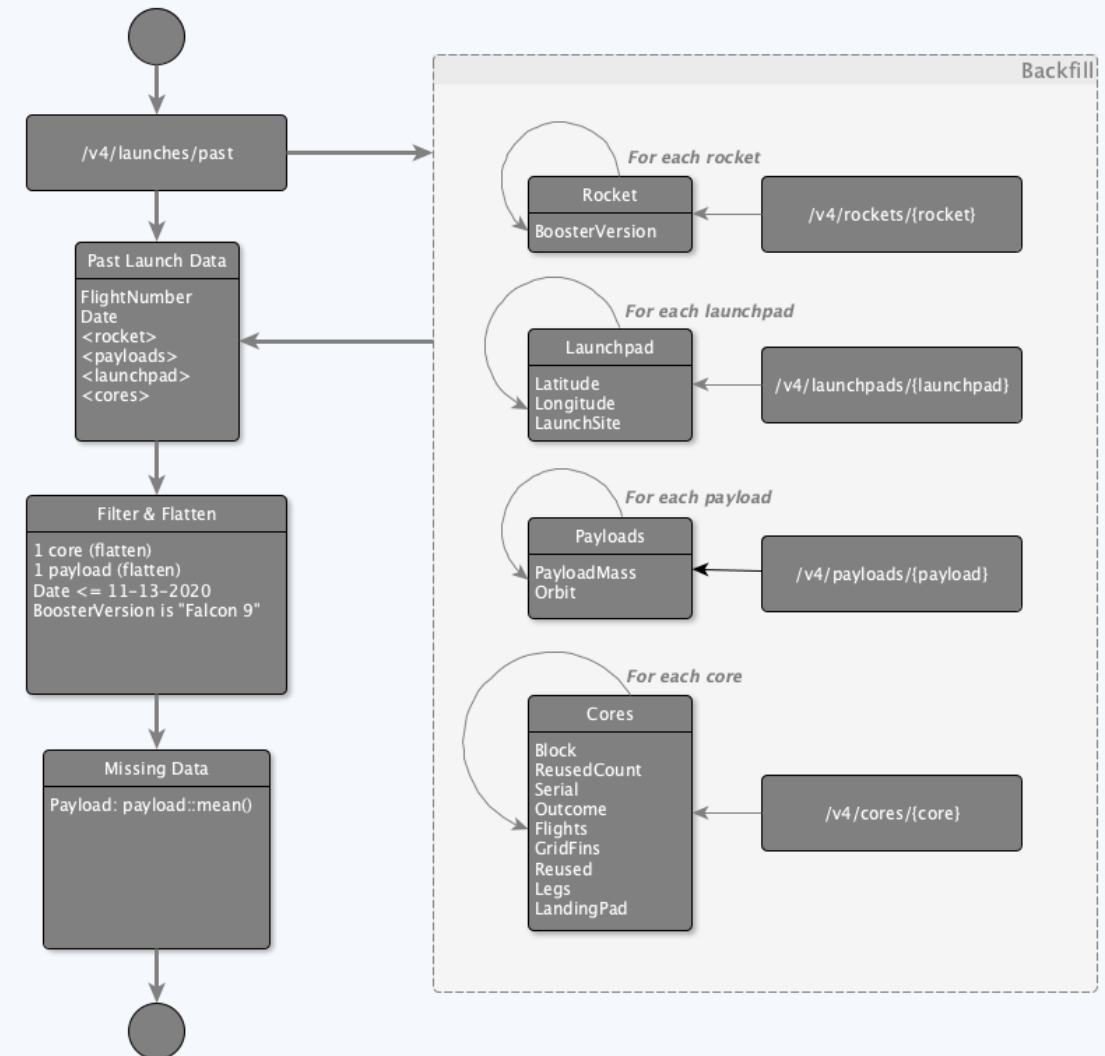
How data sets were collected:

- SpaceX REST API: GET /v4/launches/past and enrich with rockets, pads, payloads, and cores endpoints
- Parse JSON into a Pandas dataframe and keep Falcon 9 launches
- Wikipedia web scraping with BeautifulSoup for Falcon 9/Heavy tables
- Convert scraped HTML tables into a structured Pandas DataFrame

Why both sources: API provides structured flight records; scraping adds historical tabular launch details for validation and enrichment.

Notebooks: https://github.com/Itzraj786iul/Spacex_Project/blob/main/01-data-collection-api.ipynb

and https://github.com/Itzraj786iul/Spacex_Project/blob/main/01-data-collection-webscraping.ipynb



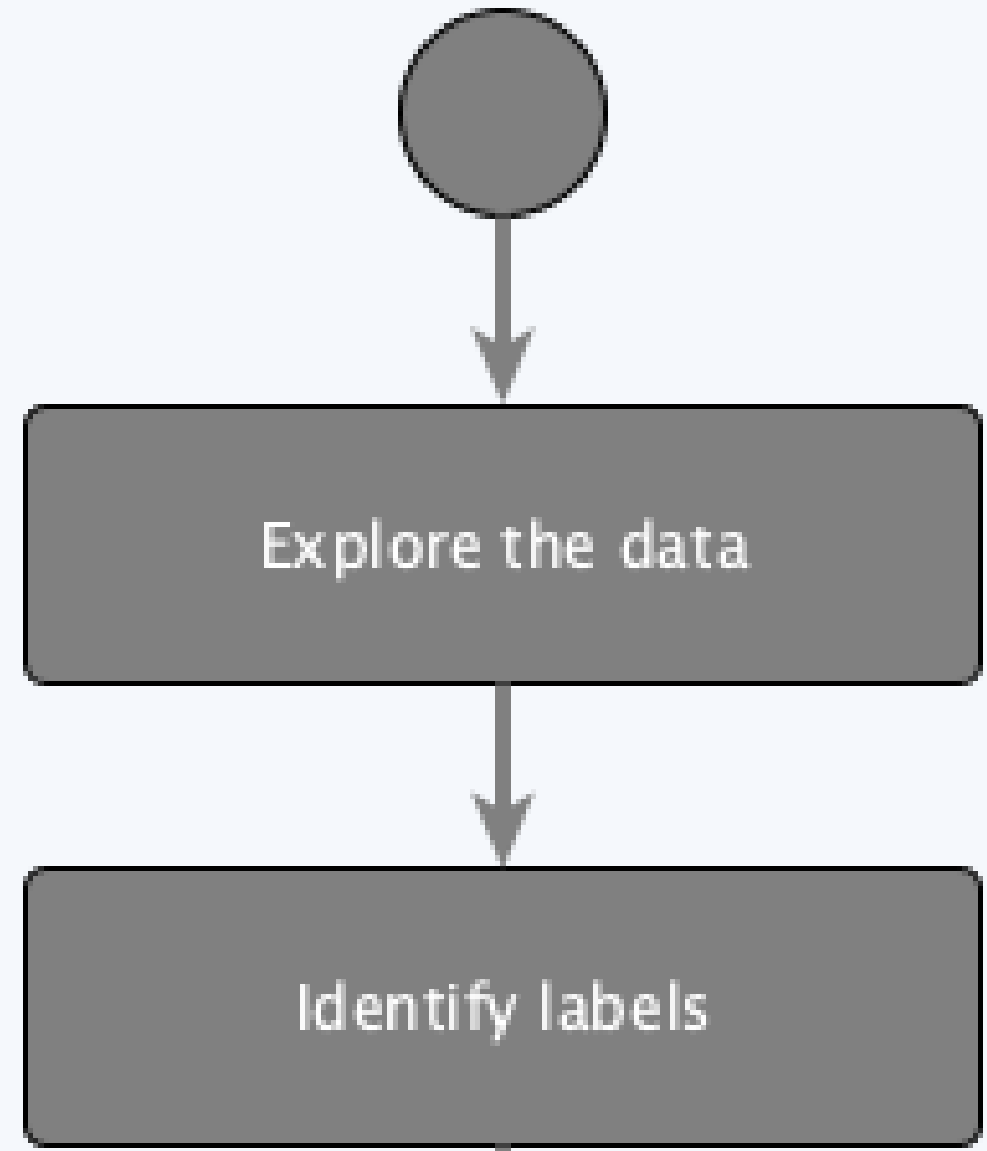
Data Wrangling Methodology

How data were processed:

- Identified missing values and column types (numeric vs categorical)
- Calculated launches per Launch Site and Orbit distributions
- Grouped landing outcomes into binary success/failure
- Created Class label: 1 = successful landing, 0 = failure
- Prepared the labeled dataset for EDA and supervised learning

Result: a clean training-ready table with outcome labels.

Notebook: https://github.com/Itzraj786iul/Spacex_Project/blob/main/01-data-wrangling.ipynb

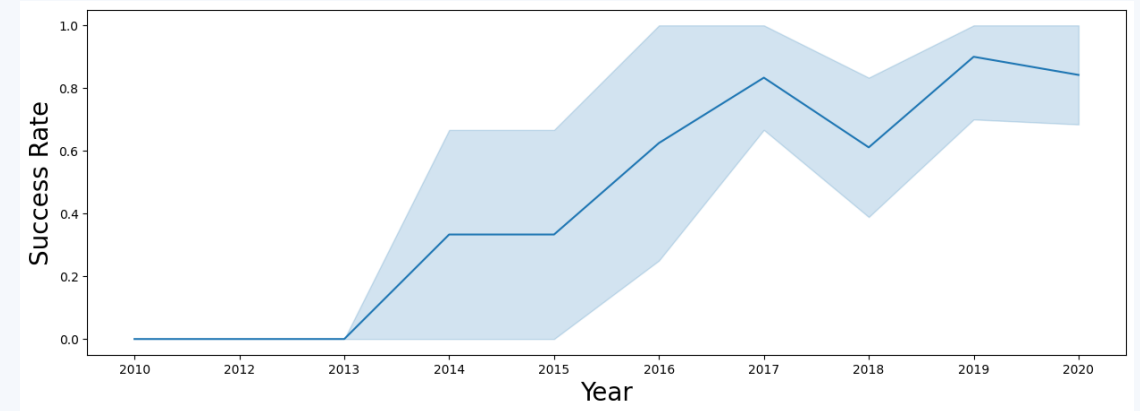


EDA and Data Visualization Methodology

Charts plotted and why:

- Flight Number vs Launch Site: success improves with experience
- Payload vs Launch Site: payload patterns differ by site
- Success Rate vs Orbit Type: compare orbit reliability
- Flight Number / Payload vs Orbit: interaction effects
- Yearly success trend: long-term operational improvement
- One-hot / dummy encoding prepared categorical features for ML

Notebook: https://github.com/Itzraj786iul/Spacex_Project/blob/main/02-eda-dataviz.ipynb



EDA with SQL Methodology

SQL queries performed (SQLite):

1. Unique launch site names
2. Five records where launch site begins with CCA
3. Total payload mass for NASA (CRS)
4. Average payload mass for F9 v1.1
5. First successful ground-pad landing date
6. Successful drone-ship landings with payload 4000-6000 kg
7. Counts of successful and failed mission outcomes
8. Booster versions that carried maximum payload
9. 2015 drone-ship failure records
10. Rank landing outcomes between 2010-06-04 and 2017-03-20

Notebook: https://github.com/ltzraj786iul/SpaceX_Project/blob/main/02-eda-sql.ipynb

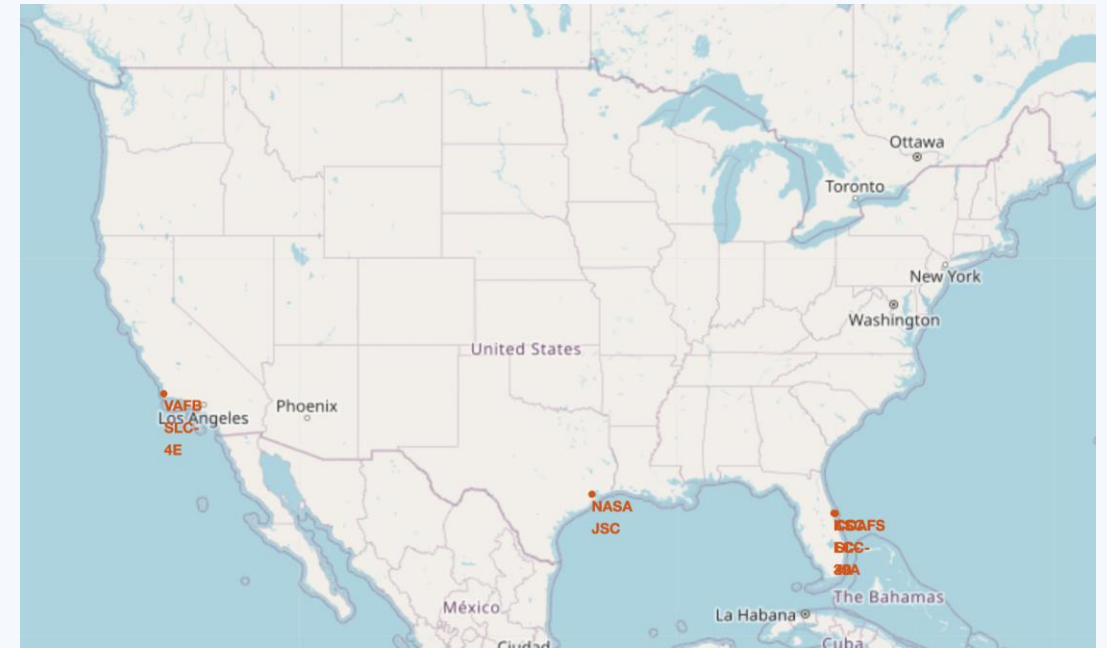
Interactive Map with Folium Methodology

Map objects created and why:

- Circles and Markers for every launch site location
- Green/red markers (and clusters) for success vs failure launches
- Distance markers and lines to railway, roadway, and coastline

Purpose: discover geographic patterns that influence site suitability and relate location to landing outcomes.

Notebook: https://github.com/Itzraj786iul/Spacex_Project/blob/main/03-folium-map.ipynb



Plotly Dash Dashboard Methodology

Dashboard plots and interactions added:

- Dropdown to select All Sites or one launch site
- Pie chart of success counts / success-failure ratio
- Payload mass range slider
- Scatter plot of Payload vs Outcome by booster version

Why: interactive filtering compares sites and payload ranges quickly.

Python file:

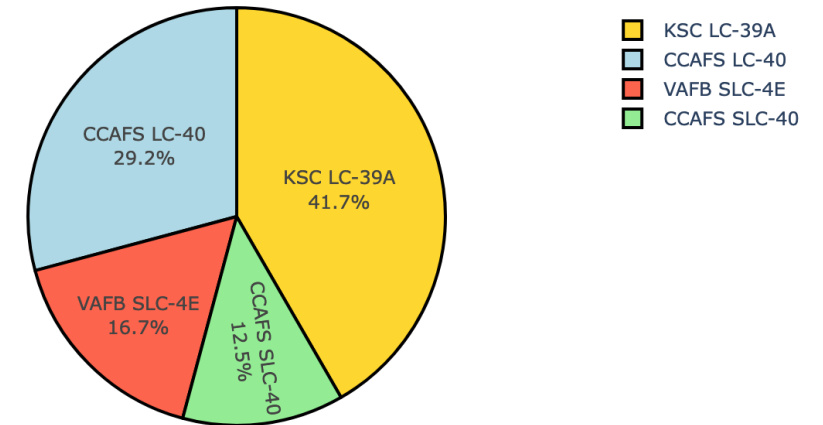
https://github.com/Itzraj786iul/Spacex_Project/blob/main/03-dash-app.py

SpaceX Launch Records Dashboard

All Sites



Total Successful Launches by Site

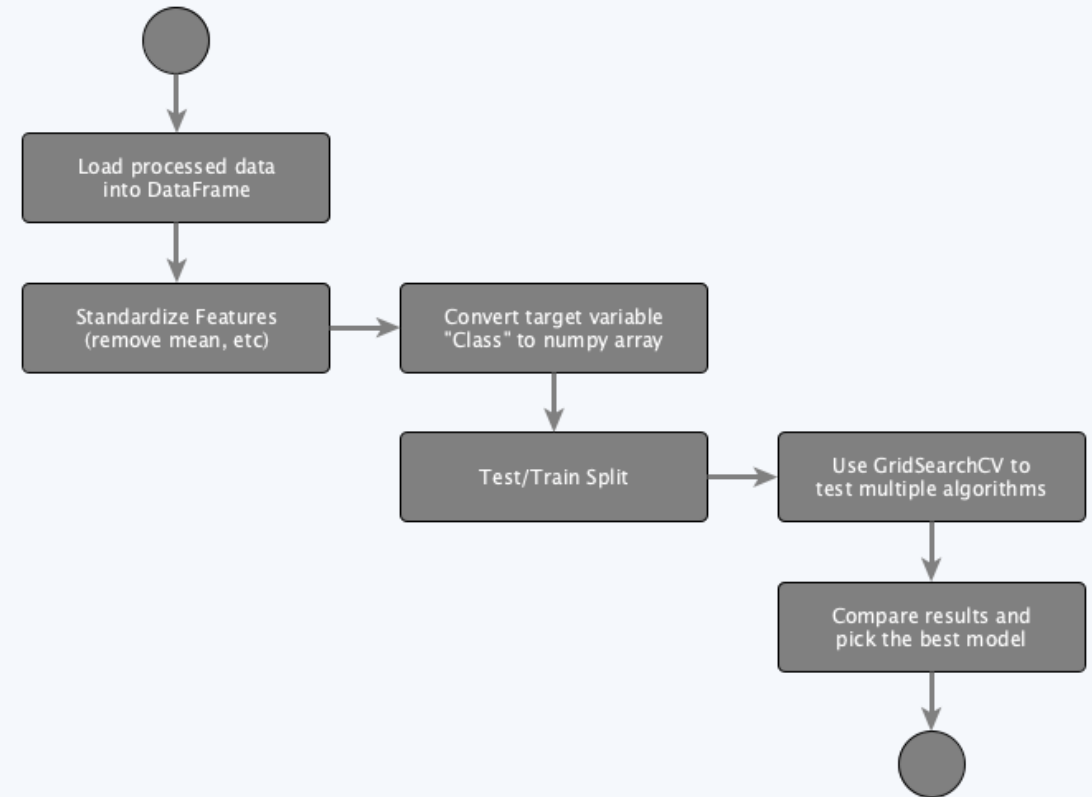


Predictive Analysis (Classification) Methodology

Model development process:

1. Standardize features with StandardScaler
2. Split train/test data (test_size=0.2, random_state=2)
3. Tune hyperparameters with GridSearchCV (cv=10)
4. Train Logistic Regression, SVM, Decision Tree, and KNN
5. Evaluate test accuracy and confusion matrices
6. Select the best performing classifier

Notebook: https://github.com/ltzraj786iul/Spacex_Project/blob/main/04-predictive-models.ipynb

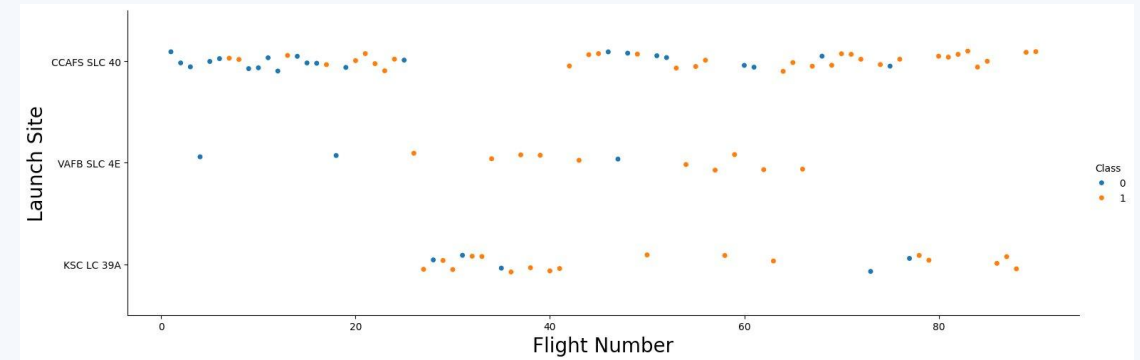


EDA with Visualization: Flight Number vs Launch Site

Scatter plot analysis:

- All sites show a mix of first-stage landing successes and failures
- Later flights generally show more successful landings
- Success improves as flight number increases

Explanation: operational experience over time improves landing success at each launch site.

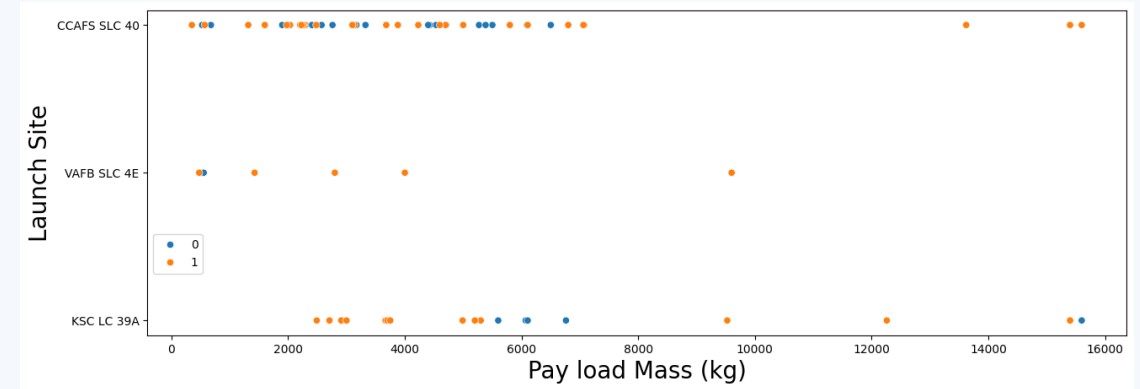


EDA with Visualization: Payload vs Launch Site

Scatter plot analysis:

- Different launch sites handle different payload profiles
- Payload mass distribution varies across CCAFS, KSC, and VAFB
- Site selection is related to mission/payload requirements

Explanation: launch site and payload mass jointly influence outcomes.

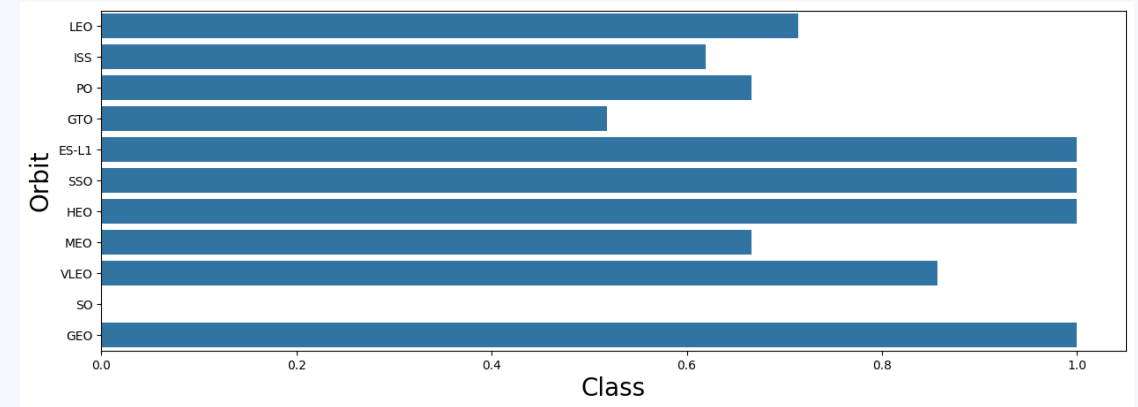


EDA with Visualization: Success Rate vs Orbit Type

Bar chart analysis:

- ES-L1, SSO, HEO, and GEO show the highest success rates
- Other orbits such as GTO show more mixed outcomes
- Orbit type is an important success-rate factor

Explanation: some orbit types are associated with more reliable recovery.

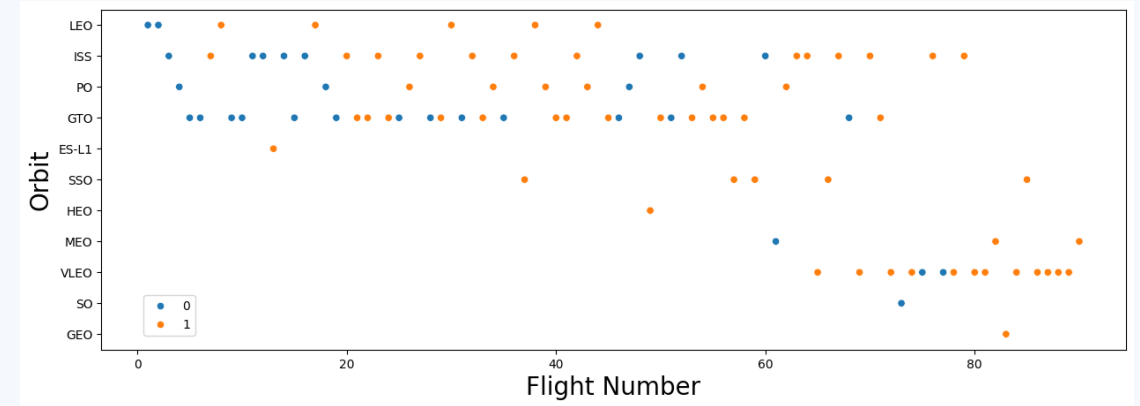


EDA with Visualization: Flight Number vs Orbit Type

Scatter plot analysis:

- Some orbits appear more frequently in later flights
- Successful landings increase for several orbits over time
- Learning effects appear across multiple orbit families

Explanation: program maturity improved outcomes across orbits.

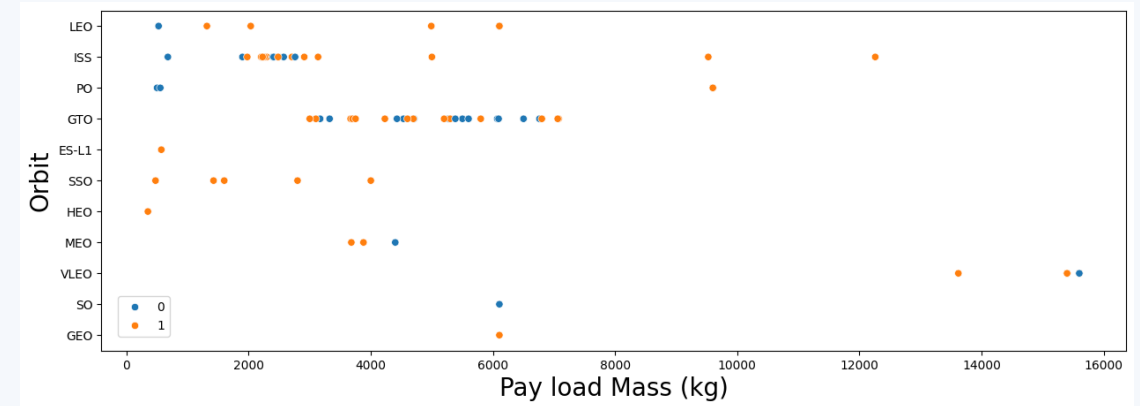


EDA with Visualization: Payload vs Orbit Type

Scatter plot analysis:

- GTO missions often carry heavier payloads with mixed outcomes
- Payload and orbit interact as predictors of landing success
- Heavy payloads on demanding orbits can reduce success probability

Explanation: payload-orbit combination is a key risk signal.

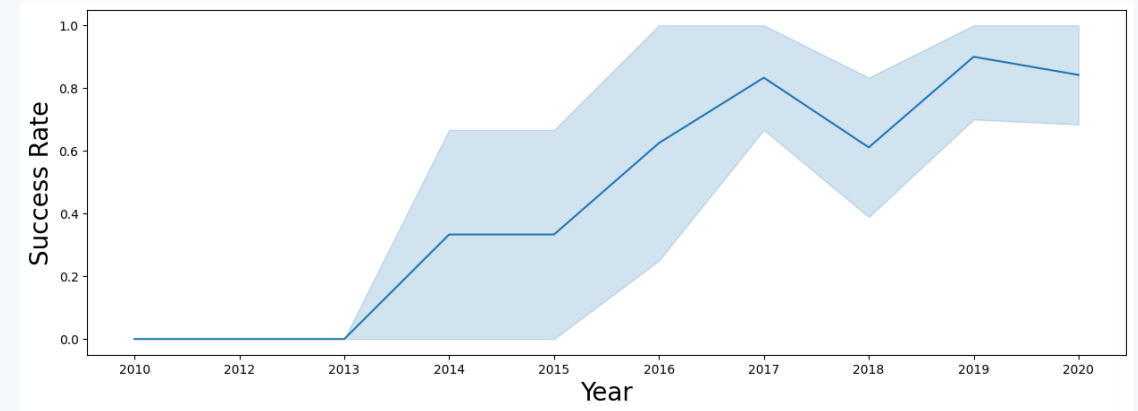


EDA with Visualization: Launch Success Yearly Trend

Line chart analysis:

- From 2016 onward, success rate improved year over year
- A minor setback appears around 2018, then recovery continues
- Recent years approach consistently high success

Explanation: Falcon 9 landing reliability improved with continuous operational and technological advancements.



EDA with SQL Results: Launch Sites and CCA Records

Query 1 - Unique launch site names:

- CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

Query 2 - Five records where launch sites begin with CCA:

- Early CCAFS LC-40 flights from 2010-2013
- Outcomes include Failure (parachute) and No attempt

Explanation: SpaceX launches from Florida and California; early CCAFS flights show recovery capability evolving over time.

EDA with SQL Results: Payload and First Ground Landing

Query 3 - Total payload mass for NASA (CRS): 45,596 kg

Query 4 - Average payload mass for F9 v1.1: 2,928.4 kg

Query 5 - First successful ground-pad landing date: 2015-12-22

Explanation:

- NASA CRS missions delivered a large cumulative payload mass
- F9 v1.1 carried moderate average payloads
- 2015-12-22 marks the first successful ground-pad landing milestone

EDA with SQL Results: Drone Ship, Outcomes, Max Payload

Query 6 - Successful drone-ship landing with payload 4000-6000 kg:

- F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Query 7 - Outcome counts show many successes and fewer failures
as the program matured (Success / Success drone ship / ground pad dominate later).

Query 8 - Maximum payload boosters are Block 5 versions
(F9 B5 B1048/B1049/B1051/B1056/B1058/B1060 family).

EDA with SQL Results: 2015 Failures and Ranked Outcomes

Query 9 - 2015 drone-ship failure records:

- January 2015 | F9 v1.1 B1012 | CCAFS LC-40
- April 2015 | F9 v1.1 B1015 | CCAFS LC-40

Query 10 - Ranked landing outcomes (2010-06-04 to 2017-03-20):

- No attempt is most frequent in this early window
- Success (drone ship) and Failure (drone ship) appear next
- Then Success (ground pad) and ocean-controlled outcomes

Explanation: early years include many no-attempt flights before reuse matured.

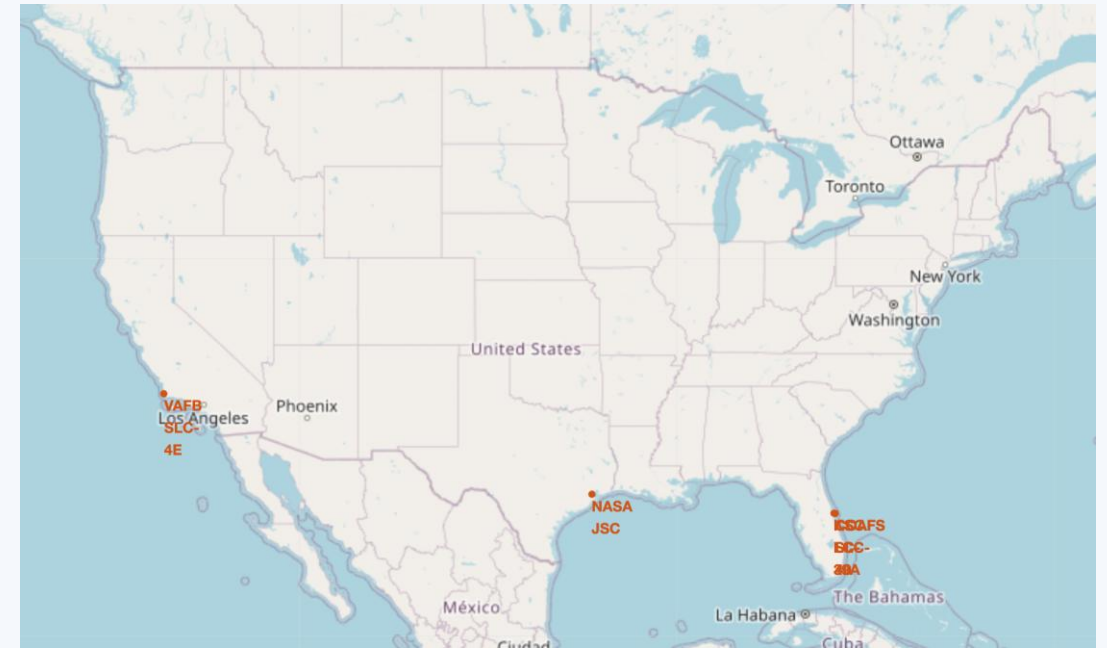
Folium Map Results: Launch Site Locations

Folium map screenshot explanation:

- All launch sites are marked near coastal regions
- Florida: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A
- California: VAFB SLC-4E

Important findings:

- Launch sites are in close proximity to the coast
- Launch sites are relatively close to the equator
- Coastal placement supports safer over-water trajectories



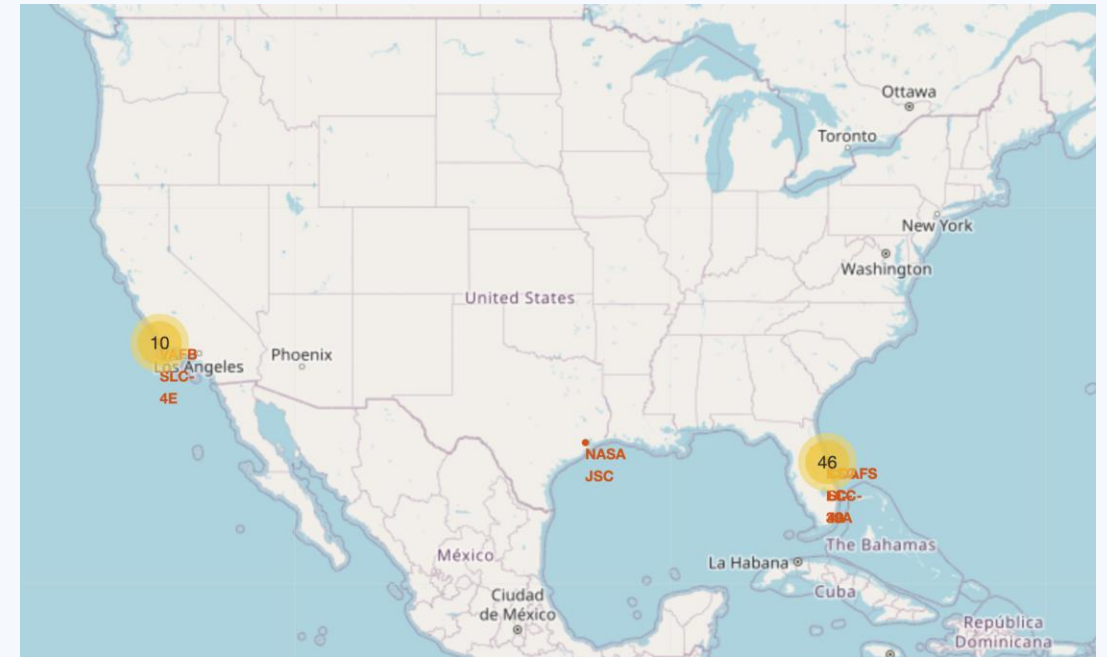
Folium Map Results: Color-Labeled Launch Outcomes

Folium map with success/failure markers:

- Green markers = successful landings
- Red markers = failed landings
- Marker clusters handle overlapping coordinates

Important findings:

- KSC LC-39A shows a high share of successful outcomes
- Color labels make high-performing sites easy to identify
- Outcome geography confirms site-level performance differences



Folium Map Results: Distances to Proximities

Distance analysis on Folium map:

- Railway about 1.02 KM from launch site
- Roadway about 0.59 KM from launch site
- Coast about 0.86 KM from launch site

Important findings:

- Sites are close to transport links for logistics
- Sites are close to coastline for flight safety corridors
- Sites keep distance from dense cities for public safety



Plotly Dash Results: Success Count for All Sites

Dashboard pie chart for all sites:

- Shows successful landing distribution by launch site
- KSC LC-39A experienced the highest proportion of successful landings

Important findings:

- Site mix is uneven; Florida sites dominate volume
- Choosing high-performing sites improves expected success

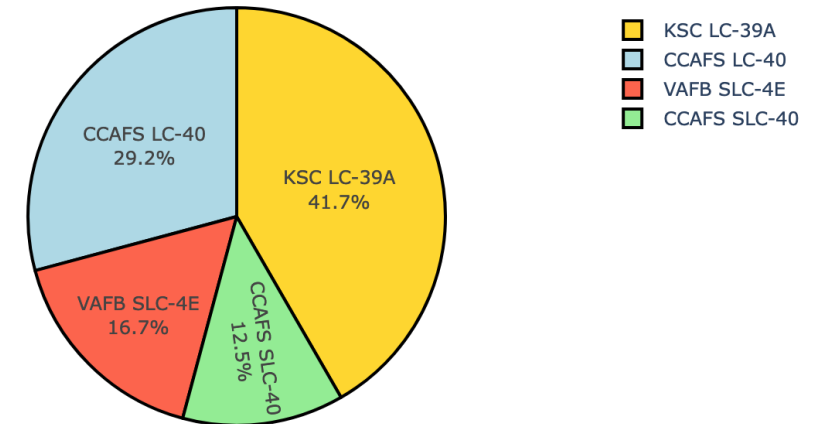
App: https://github.com/ltzraj786iul/Spacex_Project/blob/main/03-dash-app.py

SpaceX Launch Records Dashboard

All Sites



Total Successful Launches by Site



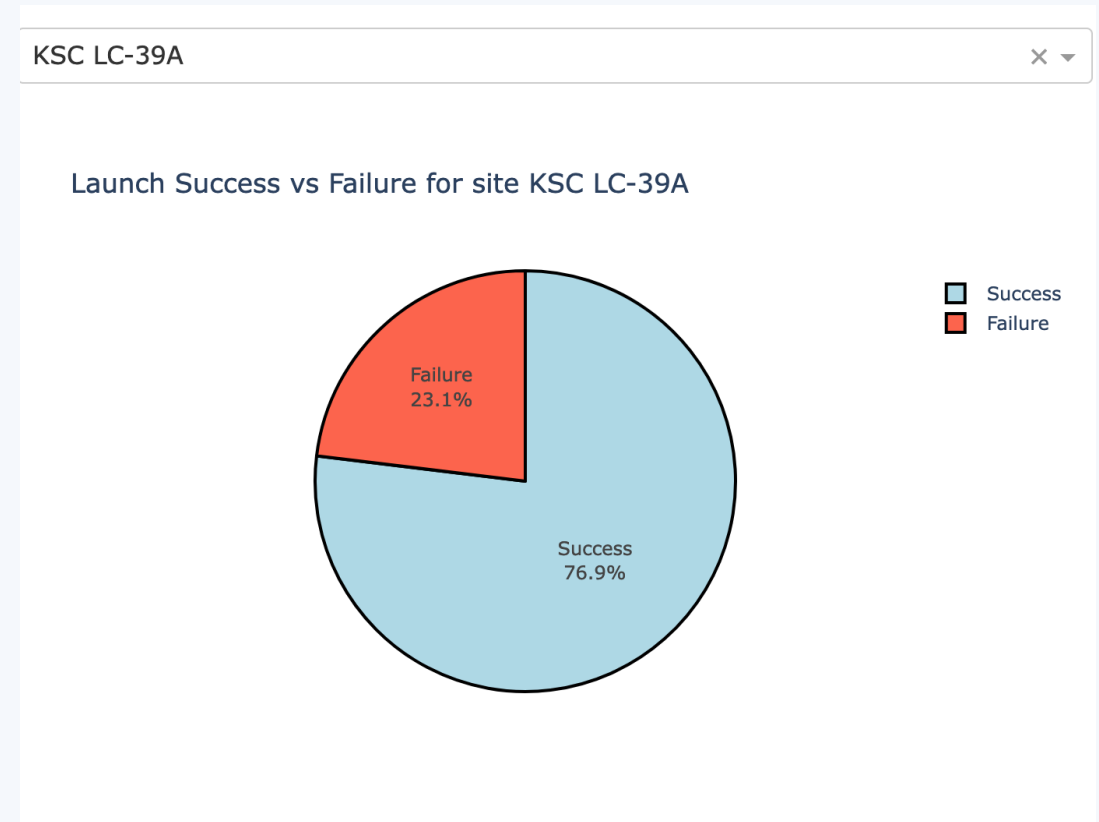
Plotly Dash Results: Highest Success Ratio Site

Dashboard pie chart for best site:

- KSC LC-39A had the highest ratio of successful landings
- Success vs Failure comparison for the selected site

Important findings:

- KSC LC-39A is the strongest site by success ratio
- Interactive site filtering confirms this consistently



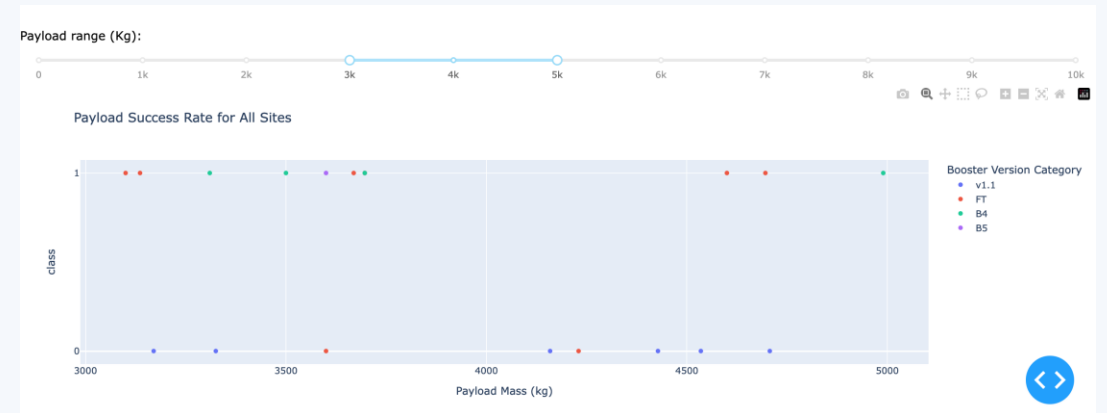
Plotly Dash Results: Payload vs Launch Outcome

Dashboard scatter with payload range slider:

- With payload mass between 3,000kg and 5,000kg, booster performance patterns become visible by version category
- Mid payload ranges often show stronger landing success

Important findings:

- Payload mass is a key interactive factor
- Booster version category helps explain outcome clusters



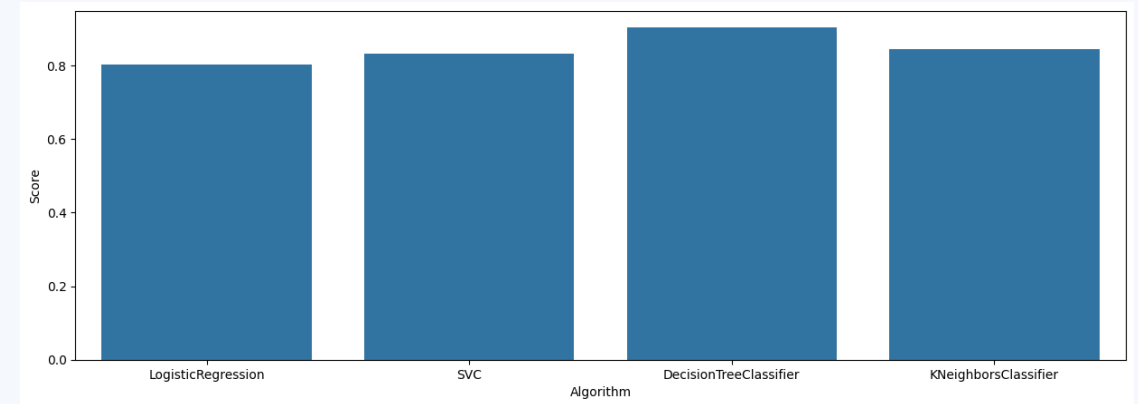
Predictive Analysis Results: Classification Accuracy

Model accuracy comparison after GridSearchCV:

- Logistic Regression: high accuracy (~0.85 validation)
- SVM: high accuracy (~0.85 validation)
- Decision Tree: highest accuracy (~0.875 validation)
- KNN: high accuracy (~0.85 validation)

Finding: DecisionTreeClassifier is the most accurate algorithm tested.

Bar chart on the right visualizes model comparison.

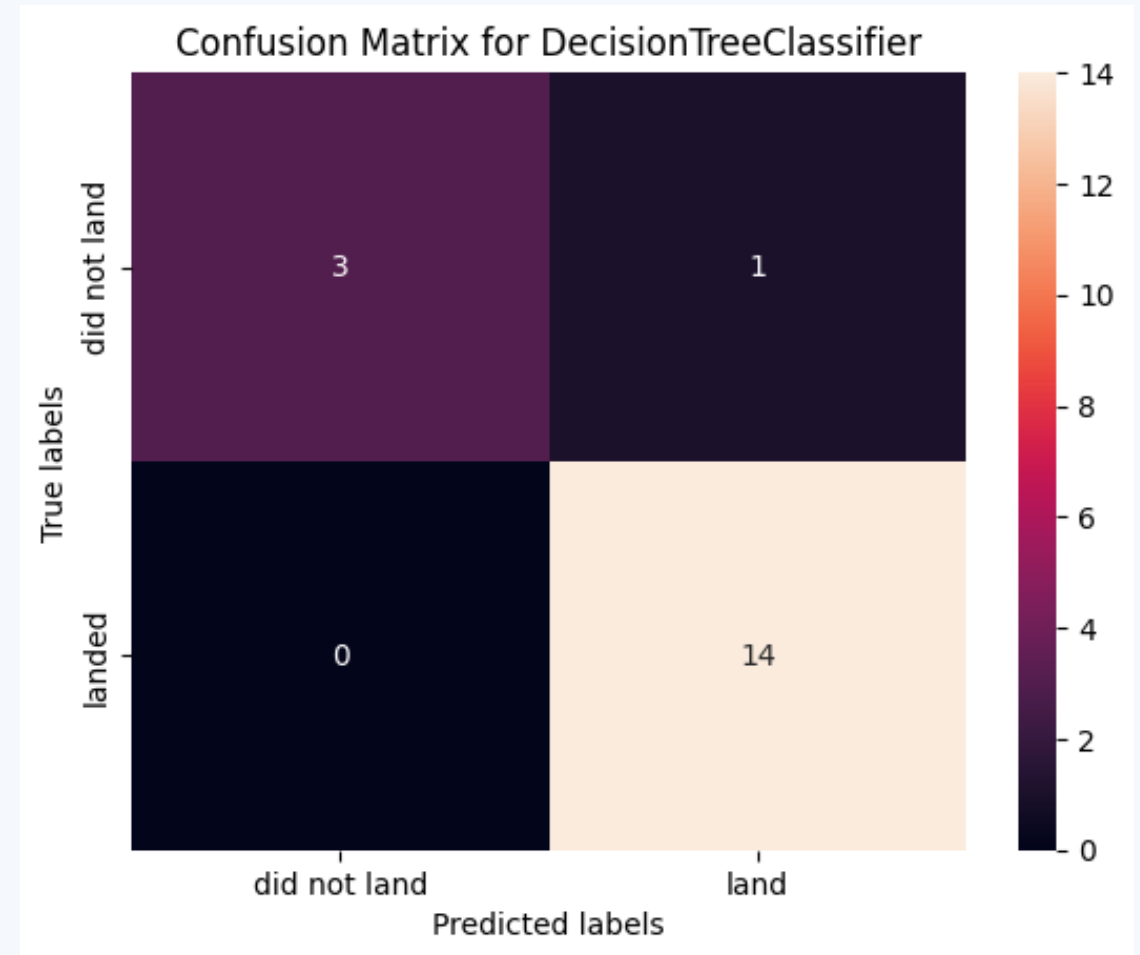


Predictive Analysis Results: Confusion Matrix

Confusion matrix of the best model (Decision Tree):

- Many observations correctly predicted as successful landings (true positives)
- Model distinguishes landed vs did-not-land classes effectively
- Remaining errors are limited on the hold-out test set

Explanation: tuned Decision Tree is selected as the final classifier for Falcon 9 first-stage landing prediction.



Conclusion

1. DecisionTreeClassifier is the best predictive model in this project
2. Launch success rates increase over time across factors
3. KSC LC-39A has the highest success rate among launch sites
4. ES-L1, SSO, HEO, and GEO orbits show consistently high success
5. Launch sites are near the coast and relatively close to the equator
6. Lower/mid payload ranges often show better landing success
7. Folium and Plotly Dash make these insights interactive and actionable

Full project repository: https://github.com/Itzraj786iul/Spacex_Project

Creativity Applied Beyond the Template

Creativity included in this presentation:

- Clear navy-branded widescreen layout for readability
- Process flow visuals for API, scraping, wrangling, and ML
- Side-by-side explanation + chart layout on results slides
- Real Folium map screenshots and Dash dashboard screenshots
- Runnable Plotly Dash app committed as 03-dash-app.py
- GitHub URL placed on title, methodology, results, and conclusion slides

1. Multi-source consensus: SQL, charts, Folium, and Dash independently identify KSC LC-39A as the top-performing site.
2. Payload-orbit risk cluster: heavy GTO-related payloads show mixed outcomes.
3. Block 5 operational edge: maximum-payload missions concentrate on B5 boosters.
4. Non-linear model advantage: Decision Tree outperforms LR/SVM/KNN, suggesting threshold interactions among site, orbit, and payload.
5. Geography supports operations: coastal + near-equator siting matches both safety corridors and observed high-success patterns.

Appendix

GitHub: https://github.com/Itzraj786iul/Spacex_Project

All completed notebooks and Python files are linked above and listed on slide 2.

Additional assets: spacex_launch_dash.csv, presentation PPTX/PDF.

How to run dashboard:

```
pip install dash pandas plotly
```

```
python 03-dash-app.py
```